



## Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

# OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

## Project Problem Statement Submission Form

### Section / Team Details

Section No.: 07

Team No.: COREOS\_OSSP

Project Title: CPU Scheduling Simulation System

### Team Members

| Roll Number | Student Name             | Signature |
|-------------|--------------------------|-----------|
| 2520030032  | A. Amulya                |           |
| 2520030275  | Kunal Shahane            |           |
| 2520030600  | Ch. Gayatri sri harshita |           |

### 1. Abstract

The CPU Scheduling Simulation System is a C-based Operating Systems project that demonstrates how a CPU manages multiple processes using different scheduling techniques. The system implements FCFS, SJF, Priority, and Round Robin scheduling algorithms and maintains process information such as arrival time, burst time, priority, and process state.

The simulator uses dynamic memory allocation, pipes, FIFOs, and signals to demonstrate process management, inter-process communication, and CPU timer interrupts. It tracks process states including Ready, Running, Waiting, and Terminated. The system calculates Waiting Time, Turnaround Time, Response Time, and CPU Utilization, and presents the results through scheduling tables and Gantt charts for easy comparison of different scheduling algorithms.

## **2. Problem Statement**

Managing multiple processes efficiently is a major responsibility of an operating system. Different CPU scheduling algorithms can significantly affect waiting time, response time, turnaround time, and CPU utilization. Therefore, a practical simulator is required to demonstrate and compare CPU scheduling techniques. The proposed system will simulate multiple processes using FCFS, SJF, Priority, and Round Robin algorithms. It will use pipes, FIFOs, signals, and dynamic memory allocation to demonstrate process communication and CPU management. The system will track process states and generate performance statistics for analyzing different scheduling approaches.

## **3. Objectives**

- To develop a C-based CPU scheduling simulation system for managing multiple processes.
- To implement and demonstrate FCFS, SJF, Priority, and Round Robin scheduling algorithms.
- To simulate process creation, execution, state transitions, and termination.
- To use pipes and FIFOs for inter-process communication.
- To simulate CPU timer interrupts and time-slice expiration using signals.
- To use dynamic memory allocation for managing process information.
- To calculate waiting time, turnaround time, response time, and CPU utilization.
- To compare the performance of different CPU scheduling algorithms using tables and Gantt charts.

## **4. Proposed Methodology**

The system will be developed in C on Linux to simulate FCFS, SJF, Priority, and Round Robin scheduling. Process information will be dynamically managed, while pipes, FIFOs, and signals will demonstrate inter-process communication and timer interrupts. Finally, scheduling performance will be evaluated using waiting time, turnaround time, response time, and CPU utilization.

## 5. Operating Systems Concepts / Linux APIs Used

| OS Concept / Linux API | Purpose in the Project                          |
|------------------------|-------------------------------------------------|
| fork()                 | Create simulated processes.                     |
| wait()                 | Synchronize process execution.                  |
| pipe()                 | Transfer process information.                   |
| FIFO                   | Enable module communication                     |
| Signals                | Simulate CPU interrupts.                        |
| Dynamic Memory         | Manage process information.                     |
| setitimer()            | Simulate time-slice expiration.                 |
| Process States         | Track process execution states.                 |
| Scheduling Algorithms  | Implement FCFS, SJF, Priority, and Round Robin. |

## 6. Individual Contribution

| Roll Number | Student Name      | Individual Responsibility                                                                                                                                                 |
|-------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2520030257  | Praneeth Gadila   | Implement process creation and management, PCB structure, dynamic memory allocation, and <b>FCFS &amp; SJF</b> scheduling algorithms.                                     |
| 2520030297  | Haadi Hasil       | Implement <b>Priority &amp; Round Robin</b> scheduling, timer-based time slicing, signals, process states, and context switching simulation.                              |
| 2520030096  | Guru Kiran Matray | Implement <b>pipes and FIFOs</b> for inter-process communication, calculate waiting/turnaround/response time and CPU utilization, and develop result tables/Gantt charts. |

## 7. Tools / Platforms / Software Used

| Tool / Platform / Software | Purpose                                                               |
|----------------------------|-----------------------------------------------------------------------|
| Linux / Ubuntu             | Development and execution environment.                                |
| C                          | Implementation of process creation, synchronization, and termination. |
| GCC                        | Compilation of C source code.                                         |
| POSIX / Linux APIs         | Process management, synchronization, and signal handling.             |
| Git / GitHub               | Version control and team collaboration.                               |

## 8. Expected Outcome

The completed system is expected to simulate multiple processes and demonstrate FCFS, SJF, Priority, and Round Robin CPU scheduling algorithms. It will use pipes, FIFOs, signals, and dynamic memory allocation to demonstrate key Linux and Operating Systems concepts. The system will track process states and generate Gantt charts and performance statistics, including waiting time, turnaround time, response time, and CPU utilization, allowing comparison of different scheduling algorithms.

## 9. GitHub Repository

GitHub Repository Link: [https://github.com/pran12171/CORE\\_OS\\_proj](https://github.com/pran12171/CORE_OS_proj)

Repository Created: \_\_\_\_\_

☐ Yes ☐ No

## Faculty Name and Signature

Faculty Name: \_\_\_\_\_

Signature: \_\_\_\_\_

Remarks: \_\_\_\_\_

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## Faculty Use

**Project Approved:** \_\_\_\_\_

☐ Yes   ☐ No

**Date:** \_\_\_\_\_

**Faculty Signature:** \_\_\_\_\_